Name: _____ **Campus ID:** _____

(26) Which of the following best defines the Turing Test, as proposed by Alan Turing in his 1950 paper "Computing Machinery and Intelligence"?

(a) The Turing Test evaluates a machine's ability to accurately compute mathematical problems without error.

(b) The Turing Test assesses whether a machine can physically resemble a human and perform tasks like a human.

(c) The Turing Test determines if a machine can engage in conversation indistinguishably from a human, such that an interrogator cannot reliably tell which is which.

(d) The Turing Test measures a machine's capacity to mimic human emotions and empathy during an interaction.

(e) The Turing Test examines whether a machine can learn and evolve over time, similar to human intelligence.

(27) Which of the following are correct limitations of a Simple Reflex Agent?

(a) A Simple Reflex Agent can only function in deterministic environments and fails in stochastic environments.

(b) A Simple Reflex Agent lacks memory, so it cannot handle partially observable environments or learn from past experiences.

(c) A Simple Reflex Agent is only capable of performing continuous tasks but fails in discrete ones.

(d) A Simple Reflex Agent requires extensive computational power to handle even simple tasks.

(e) A Simple Reflex Agent cannot adapt to new environments without any modifications to its rule set.

(28) Given a tree with a branching factor of 5 and a depth of 8 what will be the maximum number of nodes present in the tree?

(a) Above 50,000

(b) 40

(c) 488281

(d) 390625

(e) None of the above

(29) In the context of search and problem-solving, state space is defined as:

(a) Set of all possible configurations that a system or problem can be in.

(b) Set of all possible actions from a state.

(c) Number of nodes present in the tree.

(d) Number of steps needed to find the solution.

(e) None of the above.

(30) In which of the following cases Breadth First Search (BFS) will find the optimal solution

(a) If path cost is non-decreasing function of depth.

(b) If action cost are all equal to 5.

(c) If the graph is unweighted.

(d) BFS is not optimal in any case

(31) Which of the following statements about heuristics in A* search is true?

(a) A heuristic must overestimate the cost to the goal for A* to be optimal.

(b) A heuristic can be any function as long as it is non-negative.

(c) A heuristic function is said to be admissible if it never overestimates the cost of reaching the goal.

(d) The heuristic function in A* search is irrelevant to the performance of the algorithm.

(e) None of the above

(32) Consider the following search space, where A is the initial state and D is the goal states. The cost of the operator that takes you from one state to the next is given along the edge between the states, and the value of the heuristic evaluation function $h(n)$ applied to the state is written next to the node. Which of the heuristics are not admissible?

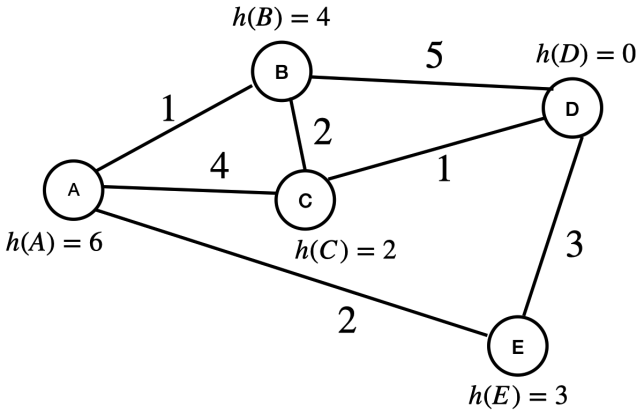


Figure 1: Caption

(a) $h(A)$

(b) $h(B)$

(c) $h(C)$

(d) $h(E)$

(e) None of the above

(33) Which of the following statements correctly defines the maximum margin classifier in the context of Support Vector Machines (SVM)?

(a) The maximum margin classifier minimizes the number of support vectors used for classification.

(b) The maximum margin classifier finds the hyperplane that separates the classes and passes through at least one data point.

(c) The maximum margin classifier finds the hyperplane that maximizes the distance to the nearest data points from each class, known as the margin.

(d) The maximum margin classifier maximizes the sum of distances between all data points and the decision boundary.

(e) The maximum margin classifier finds a hyperplane that minimizes the number of misclassified training points, regardless of the margin size.

(34) In Agglomerative clustering approach initially all points are in one huge cluster. [True/ False]

(35) Random forest solve which of the following issues with Decision Trees?

(a) Overfitting
(b) High variance
(c) Need for shallow trees for complex decision boundary
(d) Low memory footprint

(36) Which of the following is correct about Boosting?

(a) Method of combining predictions that belong to different types.
(b) Decrease variance but does not bias.
(c) All the models are influenced by the accuracy of the next model.
(d) Models are weighted according to their performance.
(e) All of the above.

(37) The limitations of the kmeans clustering are:

(a) Cannot be applied on more than 2 dimensional data.
(b) Sensitive to initializations.
(c) Can only fit as many distributions as specified by the value of k.
(d) Sensitive to outliers.

(38) Which of the following is true about Hierarchical Clustering?

(a) Very flexible in reassigning membership of data points to clusters during the clustering process
(b) Generally computationally expensive due to hierarchical nature
(c) Mistakes during clustering at one level are rectified at the next level of hierarchy
(d) None of the above

(39) Which of the following are examples of unsupervised clustering?

(a) DBSCAN
(b) K-nearest neighbour
(c) Decision Tree
(d) K-mediod
(e) All of the above

(40) Which of the following are the desired qualities of a clustering algorithm?

(a) High intra-cluster variance
(b) Low inter-cluster distance
(c) Low intra-cluster distance
(d) Homogeneity
(e) Completeness

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Fall 2024
Mid Exam – Part II – AI-500

Weightage: 60%

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Note: There are 6 questions worth 30 points. We require you to attempt all the questions.

Problem 11. (**4 pts**)

Using Alpha pruning find the list of nodes in Fig. 1 that will be pruned.

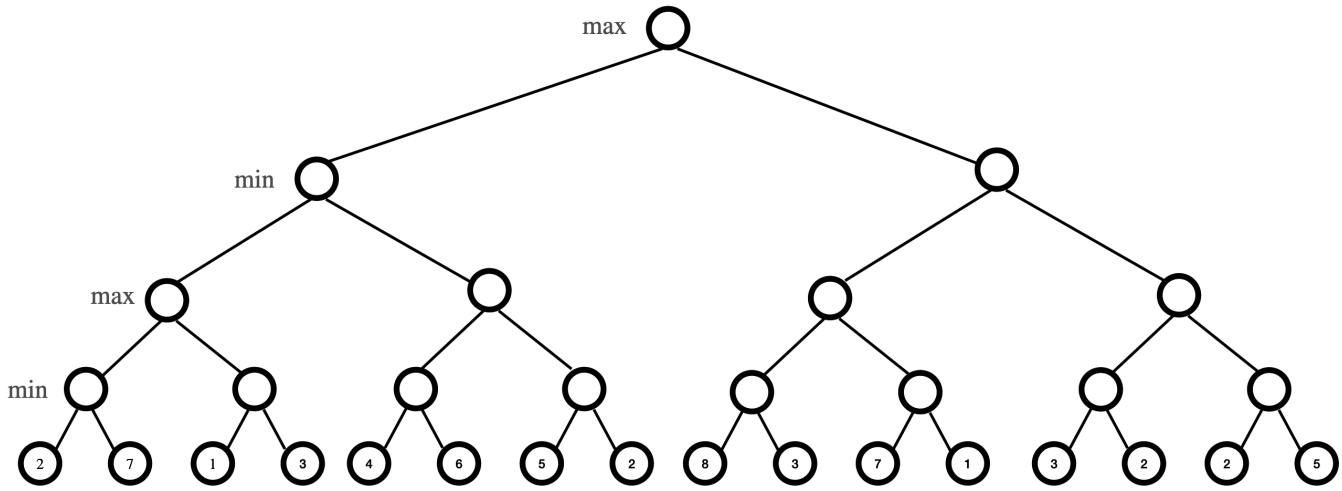


Figure 1: Alpha-Beta Pruning.

Problem 12. (**4 pts**)

Given a set of 2D points belonging to a single cluster. Assuming the point E was the mediod of the cluster in the previous iteration, find the mediod of the cluster in the current iteration using Manhattan distance.

Sr. No	Points
A	(2,3)
B	(3,4)
C	(2,5)
D	(4,1)
E	(5,2)
F	(1,3)

Mediod: _____

Problem 13. (**4 pts**)

Consider the following dataset where you want to classify a new point (5,5)

Sr. No	Points	Class
A	(2,3)	+ve
B	(3,4)	+ve
C	(2,5)	-ve
D	(4,1)	-ve

Use a weighted KNN algorithm where the weight is the inverse of the manhattan distance i.e, $\omega_i = \frac{1}{d_i}$, and $k = 7$

Problem 14. (**6 pts**)

Giving the following graph, where F is the goal node, and edge cost is defined as:

(a) $A \rightarrow B : 1$
(b) $A \rightarrow C : 1$
(c) $B \rightarrow D : 3$
(d) $D \rightarrow F : 3$
(e) $C \rightarrow E : 2$
(f) $C \rightarrow F : 5$
(g) $E \rightarrow F : 2$

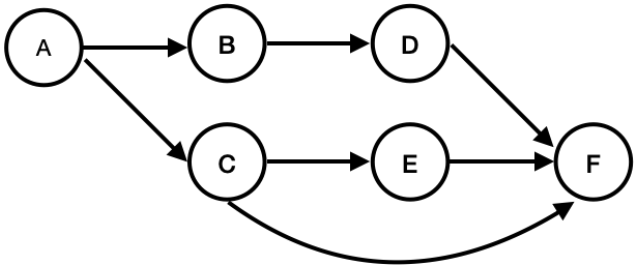


Figure 2

Find the traversal / exploration order using

(a) Breadth First Search (BFS)
(b) Uniform Cost Search (UCS)

Problem 15. (**6 pts**)

Consider the following search space, where A is the initial state and G is the goal state, where the cost of the operator that takes you from one state to the next is given along the edge between the states, and where the value of the heuristic evaluation function applied to the state is the number along the state. In cases of ties pick the state that comes earlier in the alphabet.

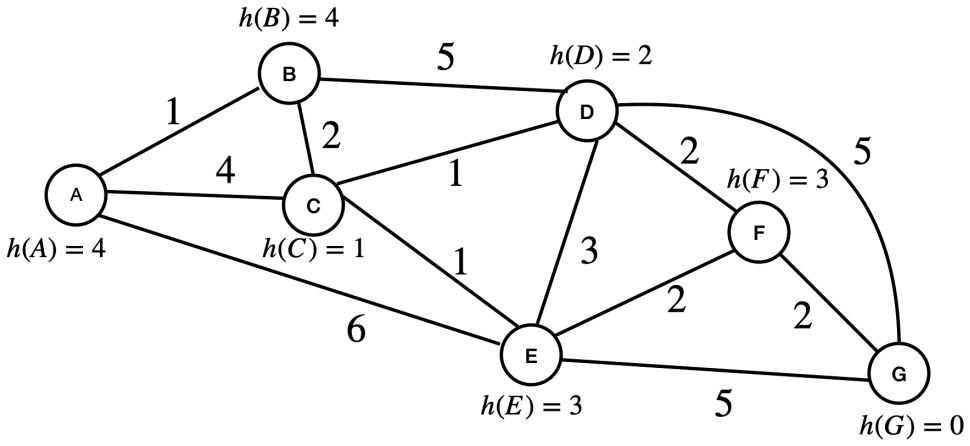


Figure 3: Heuristic Search.

Find the traversal / exploration order using

(a) Greedy Best-First Search

(b) A* Search

Problem 16. (**6 pts**) Given the following 2D points as your data (see Table 1). Considering first 6 points as your training samples, and the remaining 4 points as test set.

Sr. No	**Points**	**Labels**
1	(2,2)	+ve
2	(4,2)	+ve
3	(2,4)	+ve
4	(8,8)	-ve
5	(10,6)	-ve
6	(10,8)	-ve
7	(7,6)	+ve
8	(4,4)	+ve
9	(6,2)	-ve
10	(8,6)	-ve

Table 1: KNN Dataset

(a) Find the labels of the samples in the test set using K-Nearest Neighbour Classifier. Use Manhattan distance and $k = 3$.

(b) For the classification result in part (a), compute the precision, recall and accuracy on the test set.